

Identification of a Candidate Disease-Causing Variant in a Consanguineous Family with Intellectual Disability, Autism Spectrum Disorder, and Microcephaly Using Whole Exome Sequencing

A Whole Exome Sequencing (WES) Case Analysis Report

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Abstract

This report was prepared as part of an International Genomics Fellowship (July 1 – September 17, 2026), a collaboration between the Genomics Research Laboratory (Iran) and the Helix Institute of Biological and Research Sciences (USA), under the supervision of Dr. Sara Cheraghi.

This report presents the molecular genetic investigation of a consanguineous family (double first cousins) with two affected individuals (V:1 and V:4) presenting with developmental/intellectual delay, autism spectrum disorder, speech impairment, and progressive microcephaly (occipitofrontal circumference -2.77 SD at age 32 years in V:1, and -4.07 SD at age 30 years in V:4). Whole exome sequencing (WES) was performed and, following variant annotation and filtering, 55 candidate exonic variants across 55 genes remained. Given the consanguinity and the clinical presentation, an autosomal recessive inheritance pattern was assumed, and homozygous/hemizygous variants were prioritized. A rare, homozygous missense variant in ALKBH8 (c.1966G>T; p.D656Y), predicted damaging by multiple in silico tools (CADD = 35), emerged as the leading candidate, consistent with the previously reported association between biallelic ALKBH8 variants and autosomal recessive intellectual disability. Segregation analysis and Sanger confirmation are recommended as next steps.

1. Introduction

Intellectual disability (ID) combined with autism spectrum disorder (ASD), speech delay, and microcephaly is a clinically and genetically heterogeneous presentation. In consanguineous families, autosomal recessive (AR) inheritance is the most likely underlying mechanism, and homozygosity for rare, deleterious variants is the expected molecular signature. Whole exome sequencing (WES) has become the diagnostic method of choice for such presentations, allowing simultaneous interrogation of the coding regions of essentially all known genes.

The purpose of this report is to document the clinical and molecular work-up of a family with two affected individuals, describe the variant filtering strategy applied to WES data, and propose the most plausible causative variant based on population frequency, in silico pathogenicity predictions, ClinVar annotations, and consistency with the reported gene-disease relationship.

2. Case Presentation

This case analysis was conducted by Negin Nouri under the supervision of Dr. Sara Cheraghi, as part of an International Genomics Fellowship (July 1 – September 17, 2026) jointly run by the Genomics Research Laboratory (Iran) and the Helix Institute of Biological and Research Sciences (USA).

2.1 Family History

The family originates from a consanguineous union (the parents are double first cousins — children of two sisters). The pedigree spans five generations (I–V) and includes two affected individuals in the youngest generation: V:1 (affected male) and V:4 (affected female). All other family members are reportedly unaffected.

2.2 Clinical Phenotype

- Developmental/intellectual delay in both affected individuals
- Autism spectrum disorder (ASD)
- Speech impairment / language delay
- Progressive microcephaly:
 - V:1 — head circumference at 32 years: -2.77 SD
 - V:4 — head circumference at 30 years: -4.07 SD

Both affected individuals are the offspring of a first-cousin marriage, strongly supporting an autosomal recessive mode of inheritance.

3. Materials and Methods

3.1 Sequencing

Genomic DNA was subjected to whole exome sequencing (WES). Sequencing reads were aligned to the human reference genome and variants were called following standard best-practice pipelines (alignment, duplicate marking, base quality score recalibration, and joint genotyping).

3.2 Annotation and Filtering

Called variants were annotated using ANNOVAR against RefGene, dbSNP, 1000 Genomes, ExAC, gnomAD-derived allele frequencies, ClinVar, and a panel of in silico pathogenicity predictors (SIFT, PolyPhen-2, MutationTaster, MutationAssessor, FATHMM, PROVEAN, CADD, and others). Variants were filtered to retain rare (population allele frequency < 1%), exonic, nonsynonymous variants. Given the consanguinity and clinical presentation, particular attention was given to variants present in the homozygous (or hemizygous, for X-linked loci) state, consistent with an autosomal recessive or X-linked recessive model.

4. Results

After filtering, 55 genes harboring rare, nonsynonymous exonic variants remained. The complete list is provided in Table 1, sorted with homozygous/hemizygous variants (highlighted) first, followed by heterozygous variants ranked by CADD score.

Table 1. Candidate variants remaining after filtering (highlighted rows = homozygous/hemizygous).

Gene	Chr	Protein Change	CADD	ClinVar	Zygosity
ALKBH8	Chr 11	p.D656Y	35	Not reported	Homozygous
WDR13	Chr X	p.G206W	32	Not reported	Homozygous
DMD	Chr X	p.R814W	25.3	other Benign Benign Likely b...	Homozygous
SYNE1	Chr 6	p.K3859T	17.71	other Benign Benign	Homozygous
CDC6	Chr 17	p.D295N	34	Benign	Heterozygous
MCCC2	Chr 5	p.V339M	33	Pathogenic Pathogenic	Heterozygous

Gene	Chr	Protein Change	CADD	ClinVar	Zygosity
ABCA4	Chr 1	p.G1961E	33	other other Pathogenic Patho...	Heterozygous
CYP27A1	Chr 2	p.P384L	33	Pathogenic other	Heterozygous
GNB3	Chr 12	p.G271S	33	Not reported	Heterozygous
CHD8	Chr 14	p.R1895C	32	Not reported	Heterozygous
COL18A1	Chr 21	p.A1381T	32	Likely benign	Heterozygous
ZBTB24	Chr 6	p.R49Q	29.9	Benign	Heterozygous
SLC36A2	Chr 5	p.G87V	29.9	Pathogenic Pathogenic	Heterozygous
UNC79	Chr 14	p.L552P	29	Not reported	Heterozygous
CLCN3	Chr 4	p.G833S	28.6	Not reported	Heterozygous
EXOSC8	Chr 13	p.C89Y	28	Not reported	Heterozygous
BRAT1	Chr 7	p.L321R	27.7	Not reported	Heterozygous
COL9A3	Chr 20	p.R103W	27.5	other Benign Likely benign	Heterozygous
SLC39A5	Chr 12	p.S374T	26.4	Not reported	Heterozygous
OBSL1	Chr 2	p.R368C	26.3	Likely benign	Heterozygous
TCN2	Chr 22	p.L376S	25.6	Likely benign	Heterozygous
POP1	Chr 8	p.S568L	25.5	Not reported	Heterozygous
ZNF592	Chr 15	p.R350H	25.3	Not reported	Heterozygous
PDE6C	Chr 10	p.T696M	25.2	Not reported	Heterozygous
TMEM132D	Chr 12	p.D554G	24.8	Not reported	Heterozygous
CACNA1A	Chr 19	p.E732A	24.8	Likely benign	Heterozygous
VPS13C	Chr 15	p.R2439H	24.8	Not reported	Heterozygous
MKI67	Chr 10	p.R472W	24.5	Not reported	Heterozygous
SYNE2	Chr 14	p.D5547N	24.2	Likely benign	Heterozygous
TOPORS	Chr 9	p.R618W	24	Not reported	Heterozygous
WFS1	Chr 4	p.L432V	23.7	other Likely benign,Likely b...	Heterozygous
ATP11A	Chr 13	p.A915T	23.7	Not reported	Heterozygous
NNT	Chr 5	p.L663F	23.7	Not reported	Heterozygous
TIAM1	Chr 21	p.D1023V	23.3	Not reported	Heterozygous
SERAC1	Chr 6	p.R645C	23.2	Not reported	Heterozygous
GABRA6	Chr 5	p.Q237R	23.2	Not reported	Heterozygous
BAP1	Chr 3	p.E602V	22.9	Not reported	Heterozygous
FNDC3A	Chr 13	p.T961A	22.9	Not reported	Heterozygous
PTPRQ	Chr 12	p.Y1320C	22.8	Not reported	Heterozygous
CLCN2	Chr 3	p.D851N	22.7	Not reported	Heterozygous
CTSD	Chr 11	p.G282R	22.6	other Benign Likely benign	Heterozygous
MDH2	Chr 7	p.K259R	20.9	Not reported	Heterozygous

Gene	Chr	Protein Change	CADD	ClinVar	Zygosity
CACNA1G	Chr 17	p.R1928Q	20.8	Not reported	Heterozygous
TTC21B	Chr 2	p.R616C	20.7	Uncertain significance Benig...	Heterozygous
PAX5	Chr 9	p.A279T	20.3	not provided	Heterozygous
REEP6	Chr 19	p.R138H	20.2	Not reported	Heterozygous
TTN	Chr 2	p.D6352H	19.84	other Benign Likely benign L...	Heterozygous
NSRP1	Chr 17	p.D410N	19.78	Not reported	Heterozygous
PLEC	Chr 8	p.A1871T	18.01	Benign	Heterozygous
UBR1	Chr 15	p.S405G	16.53	Not reported	Heterozygous
RARS2	Chr 6	p.I331V	16.44	Benign Likely benign	Heterozygous
RBM28	Chr 7	p.M243L	15.71	Not reported	Heterozygous
ADD3	Chr 10	p.D532E	15.48	Not reported	Heterozygous
UNC13D	Chr 17	p.R928C	15.21	Likely benign Likely benign	Heterozygous
EPG5	Chr 18	p.N1347H	15.17	Uncertain significance	Heterozygous

4.1 Prioritization Strategy

Of the 55 candidate genes, only four carried variants in the homozygous or hemizygous state: ALKBH8, WDR13 (X-linked), DMD (X-linked), and SYNE1. Among these, the DMD and SYNE1 variants are classified as Benign/Likely benign in ClinVar and were excluded. The WDR13 variant, while rare and predicted damaging, is hemizygous on chromosome X and lacks an established gene-disease relationship with intellectual disability or microcephaly; it remains a lower-priority candidate, relevant primarily if the sequenced individual is male.

4.2 Leading Candidate: ALKBH8

- Variant: chr11, c.1966G>T (NM_138775), p.Asp656Tyr (D656Y) — homozygous
- CADD Phred score: 35 (highly deleterious)
- In silico predictions: SIFT, PolyPhen-2, MutationTaster, FATHMM, and PROVEAN all classify this substitution as damaging/deleterious
- Population frequency: rare in gnomAD/ExAC ($\approx 0.02\%$ overall)
- ClinVar: currently uncertain significance / not previously classified
- Gene function: ALKBH8 encodes a tRNA-modifying enzyme (wobble uridine hydroxylation) important for translational fidelity in neurons
- Gene-disease association: biallelic loss-of-function or damaging missense variants in ALKBH8 have been reported in autosomal recessive intellectual disability (MRT71), often accompanied by speech delay and, in a subset of patients, microcephaly

5. Discussion

The homozygous ALKBH8 variant identified in this family is the strongest candidate to explain the observed phenotype for several converging reasons: (1) it is compatible with the autosomal recessive inheritance pattern expected in a consanguineous pedigree; (2) it is rare in population databases; (3) it is predicted deleterious by essentially all computational pathogenicity tools; and (4) ALKBH8 has a previously reported, phenotypically consistent association with intellectual disability, speech delay, and, in some patients, microcephaly.

Other heterozygous variants identified in the filtered list — including well-known neurodevelopmental genes such as CHD8 — do not fit the recessive inheritance model implied by the consanguinity and by the fact that two siblings are affected while parents are presumably unaffected carriers. CHD8-related disorders, in particular, are typically caused by de novo dominant loss-of-function variants and are often associated with macrocephaly rather than microcephaly, which further argues against CHD8 as the causal gene in this family.

5.1 Limitations

- The candidate variant has not yet been confirmed by Sanger sequencing
- Segregation analysis in additional family members (parents as obligate carriers, unaffected siblings) has not been performed

- WES data from the second affected individual (V:4) was not available for direct comparison at the time of this analysis
- Functional/biochemical validation of the ALKBH8 p.D656Y variant has not been performed

6. Conclusion and Recommendations

A rare, homozygous, predicted-damaging missense variant in ALKBH8 (p.D656Y) is proposed as the leading candidate for the phenotype observed in this consanguineous family. The following steps are recommended to confirm this finding:

- Sanger sequencing confirmation of the ALKBH8 variant in the proband
- Segregation analysis across the family (parents, unaffected siblings, and the second affected individual V:4)
- Re-sequencing or targeted analysis of V:4 to confirm the presence of the same homozygous variant
- Literature review and, if feasible, functional studies to further support pathogenicity
- Genetic counseling for the family regarding autosomal recessive inheritance and recurrence risk

7. References

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